

MULTI-LEVEL CONVERSION: HIGH VOLTAGE CHOPPERS AND VOLTAGE-SOURCE INVERTERS

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KEY-WORDS

Static converters, high voltage, high frequency, series connection of semiconductors, multilevel converters.

ABSTRACT

This paper is focused on high voltage power conversion. Conventional series connection and three-level voltage source inverter techniques are reviewed and compared. A new versatile multilevel commutation cell is introduced; it is shown that this new topology is safer, more simple to control, and delivers purer output waveforms. The authors show how this technique can be applied to either choppers or voltage-source inverters and generalized to any number of switches.

INTRODUCTION

In the field of High Voltage Power Conversion, the circuit designer is often confronted to a serious problem: there are no semiconductors capable of sustaining the desired voltage (traction application for example). For this reason, circuit designers proposed several converter topologies in which only a fraction of the voltage is applied to each switch.

1 SERIES CONNECTION OF SEMICONDUCTORS

Circuit

The first solution involves series connection of several switches controlled in synchronism, thus obtaining the equivalent of a high voltage switch (Fig. 1).

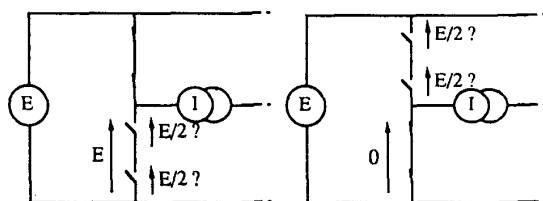


Figure 1
Plain series connection of switches

Topology

Any topology of converter (chopper, inverter, ...) can use such a commutation cell.

Voltage sharing

Static and dynamic sharing of the voltage across the switches is quite difficult to obtain and requires specific techniques:
- static balancing can be simply achieved by connecting large resistors in parallel with each switch,
- dynamic balancing is a more serious problem. The designer must make sure that all the switches commute at the very same instant: otherwise the switch that turns off first (or that turns on last) would have to sustain all of the voltage.

Control

In most cases, synchronizing the switchings, cannot be obtained by simply synchronizing the control signals; selecting semiconductors with paired turn-on and turn-off delays or using control circuits capable of compensating for the turn-on and turn-off delays is generally required.

dV/dt

With this technique, everything is done to contrive all the switches to commute at the same time. When this is achieved, the dV/dt generated at each commutation is the sum of the dV/dt generated by all the switches. Such dV/dts will induce important noise that can be dangerous for the surrounding low-level circuits and especially for the firing circuits of the high-side switches.

Voltage levels

From a control viewpoint, although a greater number of switches is used, no extra degree of freedom is gained; the series-connected switches MUST behave as one single switch and the commutation cell delivers either 0 or E (Figure 1).

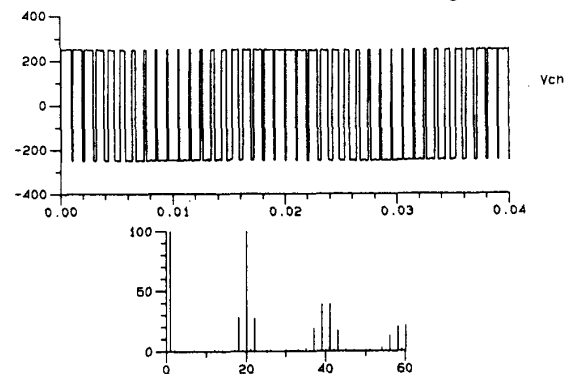


Figure 2
Waveforms and harmonic spectrum obtained with an inverter using series connection of semiconductors.

Harmonic spectrum

As a consequence, the amplitude of the harmonic at the switching frequency is very high (Fig. 2).

Extension to n switches

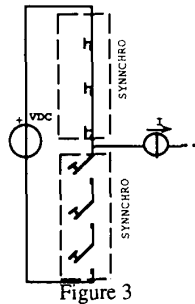


Figure 3

Increasing the number of switches connected in series (Fig. 3) does not fundamentally change the problem. However, in practice, the dV/dt and synchronization problems become even worse.

2 CONVENTIONAL THREE-LEVEL INVERTERS

Circuit

The circuit of the three-level inverter uses intermediate voltage sources and extra diodes (Figure 4).

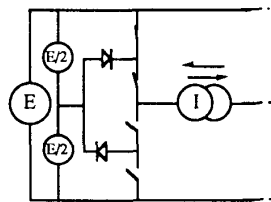


Figure 4

Three-level inverter leg

Voltage sharing

The three-level inverter leg can first be seen as a commutation cell using series-connected semiconductors in the which clamping circuits ensure the voltage sharing instead of forcing the switches to commute at the very same time. However, it should be noted that only the "outer" switches are efficiently protected by the clamping diodes (Fig. 5).

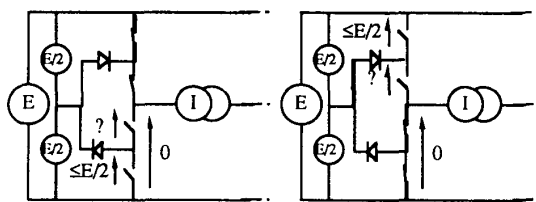


Figure 5

Voltage sharing in a three-level inverter

dV/dt

Since the commutations of the different switches need not occur at the same time, the dV/dt can stay at a reasonable value.

Voltage levels

In fact, the main advantage of this circuit is that the clamping circuits allow using a new combination of conducting-blocking switches (Fig. 6): turning the "inner" switches on and the "outer" switches off connects the current source to the intermediate voltage level $E/2$. Because of this extra level, this topology is generally referred to as "3-level" (0, $E/2$, E) converter.

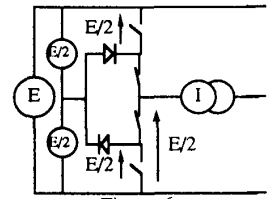


Figure 6

Intermediate level delivered by the three-level commutation cell

Topology

In practice, the two voltage sources $E/2$ are two capacitors charged at $E/2$. This is equivalent provided the average current in each capacitor is zero. Each time the current source I is connected to an intermediate voltage source, current I flows through the capacitors of the voltage dividers. If current I is unidirectional, the current in the capacitors are unidirectional and their voltages cannot stay at the desired value. As a conclusion, chopper operation is not possible with such converters.

This also shows that in a three-level voltage-source inverter, the values of the capacitors in the voltage dividers are imposed by the frequency of the current source, and is not related to the switching frequency, which leads to high value capacitors.

However, this last remark is not true when several inverter legs use common intermediate voltage sources and when the low frequency components of the capacitor currents cancel each other (for example, in a three-phase inverter with balanced load, the capacitors must only filter the switching frequency).

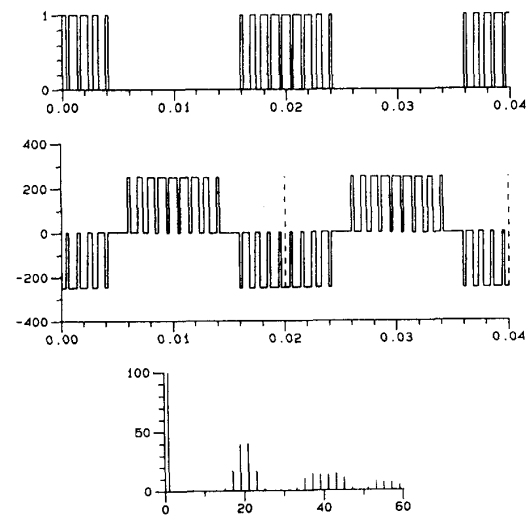


Figure 7

Waveforms and harmonic spectrum of a conventional three-level inverter.

Control

Of course, it is advisable to take benefit of the extra voltage level to improve the harmonic spectrum of the output voltage. In recent years, several studies dealing with the control of such converters have been carried out and many strategies have been proposed. One of them (ref. 1) allows modulating the output voltage between 0 and $E/2$ during one half of the period, and between $E/2$ and E during the other half of the period. With this control strategy, modulation of the output voltage is achieved by one pair of switches for one half-period and by the other pair of switches for the other half-period.

Harmonic spectrum

From an harmonic viewpoint, this control roughly leads to a reduction of the amplitude of the harmonic at the switching frequency by a factor of two (Fig. 7).

Extension to n switches

Of course, the circuit can be generalized to greater number of switches (Fig. 8), but there might be still a lot of work to do to optimize the control of such a converter?

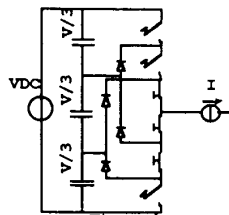


Figure 8
Four-level inverter leg

3 VERSATILE MULTILEVEL COMMUTATION CELL

Circuit

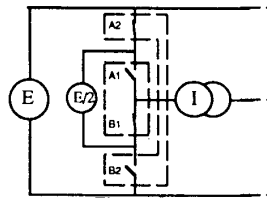


Figure 9
Versatile multilevel commutation cell

The versatile multilevel commutation cell is shown in Fig. 9. The switches are arranged in two pairs (A_1, B_1) and (A_2, B_2). Within each pair, the switches obey the same rule as the two switches of a conventional commutation cell: they must ALWAYS be in complementary states.

Voltage sharing

The voltage across the blocking switch of any pair is imposed by voltage sources E and $E/2$. It can be seen from figure 10 that the voltage across any blocking switch is always $E/2$.

Hence, this topology solves the problem of static and dynamic sharing of the voltage across the blocking switches.

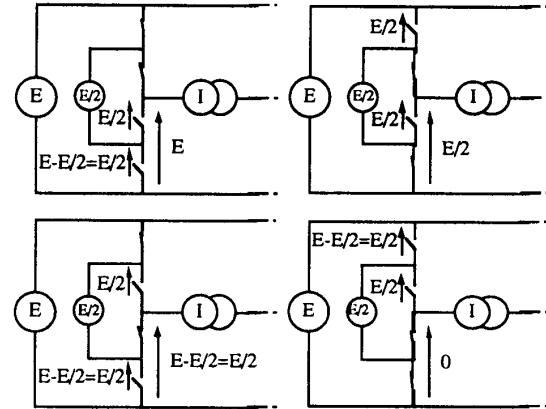


Figure 10
Voltage sharing in the new multilevel commutation cell

Hence, this topology solves the problem of static and dynamic sharing of the voltage across the blocking switches.

dV/dt

In this circuit, the switches of the different pairs can be controlled at different times, which allows limiting the dV/dt to standard values.

Voltage levels

Since the voltage across a blocking switch is $E/2$ (and assuming that the voltage across a conducting switch is zero) the voltage delivered by this commutation cell can be 0, $E/2$, or E (like in a conventional multilevel voltage-source inverter), depending on whether there is 0, 1 or 2 blocking switches among B_1 and B_2 .

Topology

In practice, the floating source $E/2$ will have to be replaced with a capacitor charged at $E/2$. Does this result, like in the conventional three-level inverter, in some limitation of the field of application of this circuit?

It can be seen that, depending on the states of the switches, the current flowing through capacitor C is $-I$ (B_1 and A_2 conducting), 0 (A_1 and A_2 conducting, OR B_1 and B_2 conducting) or $+I$ (A_1 and B_2 conducting). So, it can be seen that the capacitor current can be directly modulated by controlling the switches in an appropriate manner. We will show later on that this property allows keeping the capacitor voltage stable in choppers as well as in inverters.

Generalization to n switches

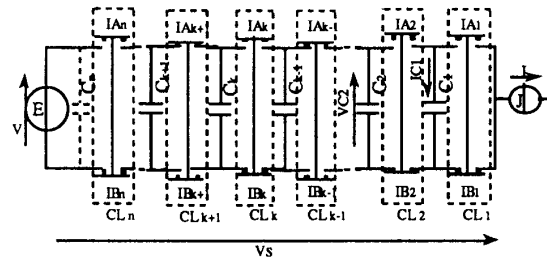


Figure 11
Multilevel commutation cell

As a first approach, we will admit that the capacitor voltages are as follows:

$$V_{ck}=k.V/n \quad k=1,\dots,n$$

The voltage across the blocking switch of pair #k only depends on the voltage of capacitors C_k and C_{k-1} and is given by:

$$V_{offk}=k.V/n-(k-1).V/n=V/n.$$

Knowing that the voltage across a blocking switch is V/n (and assuming that the voltage across a conducting switch is zero) gives a very straightforward understanding of the operation of the converter: to determine the voltage delivered by the multilevel commutation cell at any time, count the number of blocking switches on the low side and multiply by V/n .

This shows that there are $n+1$ possible levels: 0, V/n , $2.V/n, \dots, V$ (like in a conventional multilevel voltage-source inverter).

Control

The control of the multilevel commutation cell must fulfil simultaneously two important requirements:

- 1 - compatibility with $V_{ck}=k.V/n=\text{constant} \quad k=1,\dots,n$
- 2 - optimization of the harmonic spectrum

To determine the appropriate control pattern for the n pairs of switches, we will now assume that the initial value of voltages V_{ck} are given by $V_{ck}=k.V/n$ for $k=1,\dots,n$ and study the conditions to meet in order to keep these voltages constant.

Each capacitor C_k is connected between the pairs of switches #k and #k+1. Depending on the state of these pairs, the current in the capacitor is +I, 0 or -I. It can be expressed as follows:

$$i_{ck}=(s_k-s_{k+1}).I$$

in which s_k and s_{k+1} are 0 or 1, depending on the state of the corresponding pair of switches ($s_k=0$ when switch A is conducting and $s_k=1$ when switch B is conducting)

This equation gives the following steady-state stability condition for voltage $V_{ck} \quad k=1 \dots n$

$$(i_{ck})_{av}=0 \rightarrow \int_0^T I(t).(s_k-s_{k+1}).dt=0$$

When I is quasi constant for a switching period (which includes choppers but also PWM inverters with $f_{\text{switching}} \gg f_{\text{modulation}}$), the steady state stability condition for voltage $V_{ck} \quad (k=1 \dots n)$ can be written:

$$(i_{ck})_{av}=0 \rightarrow \int_0^T I_{DC}.(s_k-s_{k+1}).dt=0 \rightarrow \int_0^T (s_k-s_{k+1}).dt=0$$

The trivial case where all the control signals are the same corresponds to the conventional series connection technique. If all the switches are controlled at the same time, the current in the capacitors is theoretically always zero and the capacitors are not needed.

More generally, the stability condition is satisfied as soon as the control signals have the same duty cycle, whatever their relative phases. Hence, as soon as n phase-shifted signals are used to control these multilevel converters, the stability condition for capacitors $C_1, \dots, C_n$ is satisfied.

We will now see that the value of the phase-shift has a strong influence on the harmonic spectrum of the output voltage.

When the phase-shift is very small, the output voltage is very close to that delivered by a conventional chopper, except that the voltage steps are composed of n voltage steps of amplitude V/n . In this case, the RMS current in capacitors $C_k \quad k=1, \dots, n$ is very small and the capacitors can have a very low value.

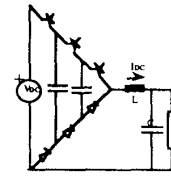
However, it is also possible to improve the harmonic spectrum of the output voltage by using greater phase-shifts. Since the voltage only depends on the number of conducting diodes, using a set of control signals phase-shifted of T/n produces an output voltage which is periodical with a period T/n . This shows that all the harmonics up to n times the switching frequency are zero. With such control signals, the

ripple of the output voltage is only V/n whatever the point of operation. So, the amplitude of the first non-zero harmonic is roughly n times smaller than the harmonic at the switching frequency obtained with a converter using plain series connection of switches. It can be seen that this is probably a pessimistic approximation because when the conduction time of the transistors is a multiple of T/n , the output voltage ripple is theoretically zero (as well as all the harmonics).

3.1 Application to a multilevel buck converter

Like in conventional choppers, the study of the static characteristics of the switches shows that they must be unidirectional for voltage and current, and the study of the dynamic characteristics of the switches show that "A" switches must be turn-on and turn-off controlled and that "B" switches must commute spontaneously. In summary, when IDC is a positive current source "A" switches must be bipolar, MOS, IGBT or GTO and "B" switches must be diodes.

Figure 12 shows the simulation results obtained with a three transistors-three diodes step-down chopper.



Four-level buck converter

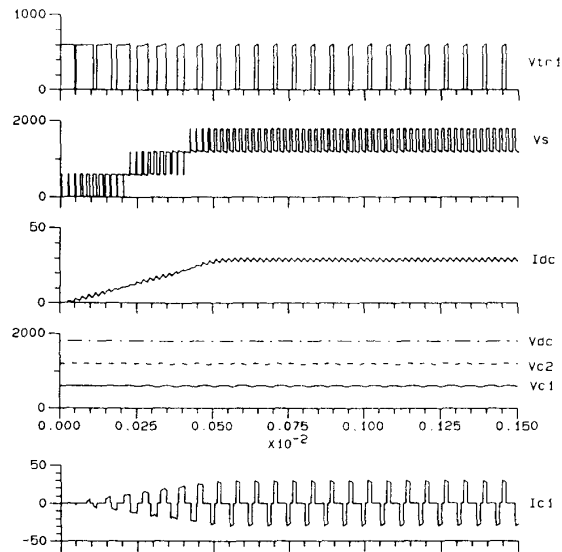


Figure 12
Operation of a four-level buck converter

Stability of the voltage distribution

Up to now, we have always supposed that the initial voltage distribution was correct. The stability of the $V_{ck}=k.V/n$ distribution can be demonstrated as soon as the current source has a finite impedance (Ref 2). In the interest of brevity, we will only show in this paper two simulations carried out with wrong initial conditions (Fig. 13). It can be seen that the open-loop system naturally evolves towards the desired voltage distribution.

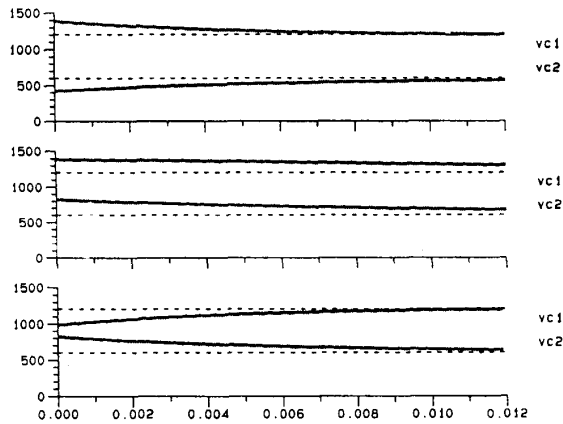
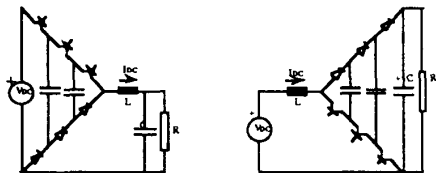


Figure 13
Natural balancing of the capacitor voltages

3.2 Application to other choppers

Other multilevel topologies are derived from the multilevel commutation cell like conventional topologies are derived from the conventional commutation cell (ref. 3): buck, boost, buck-boost, Cuk... are obtained by connecting the second terminal of the current source at different voltage levels (Fig. 14).



Four-level buck converter Four-level boost converter
Figure 14
Multilevel choppers

3.3 Application to a PWM three-level inverter in half-bridge configuration

Like in conventional voltage source inverters, the bidirectional current source requires reverse conducting devices, and each switch must be composed of a bipolar transistor, MOSFET, IGBT, GTO... with an antiparallel diode (Fig. 15).

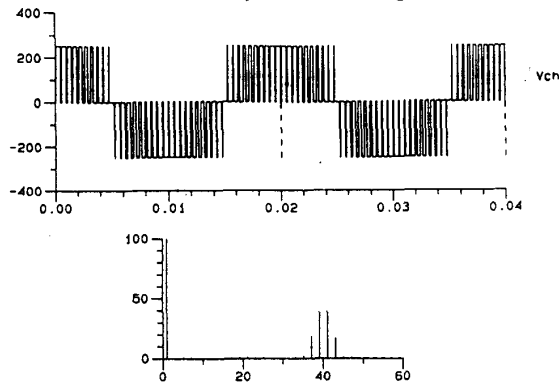
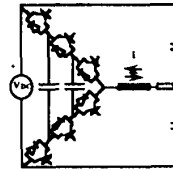


Figure 15
Waveforms and harmonic spectrum obtained with the new three-level inverter in half-bridge configuration

The waveforms and harmonic spectrum obtained with this converter in three-level configuration (Figure 15) can be compared to those obtained with plain series connection and conventional multilevel inverter

3.4 Application to other PWM inverters

Figure 16 displays the results obtained with a six transistor-six diodes voltage-source inverter in half-bridge configuration.



Half-bridge 4-level inverter

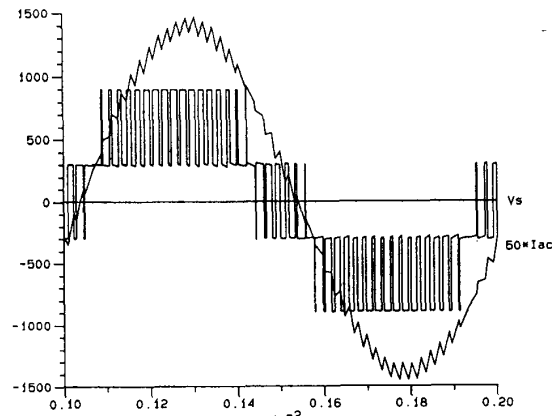


Figure 16
Waveforms of a 4-level half-bridge inverter

This multilevel inverter leg offers the same versatility as the conventional inverter leg:

- any voltage-current phase-shift between 0° and 360° is attainable which means, in particular, that power can flow from the DC voltage source to the AC current source or in the opposite direction (figure 17)

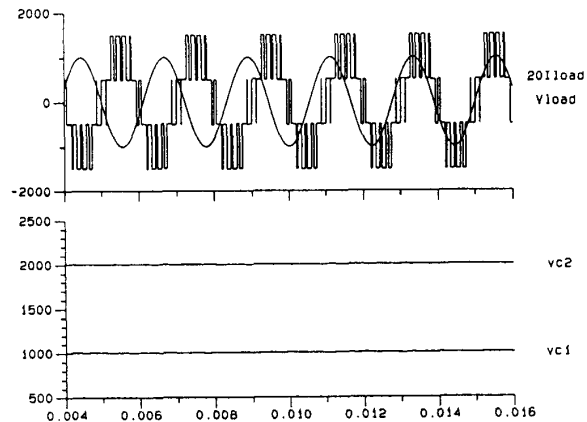


Figure 17
Variable phase shift operation.

- "over-modulation" is directly possible since the 0 and V levels do not involve the capacitors (figure 18).

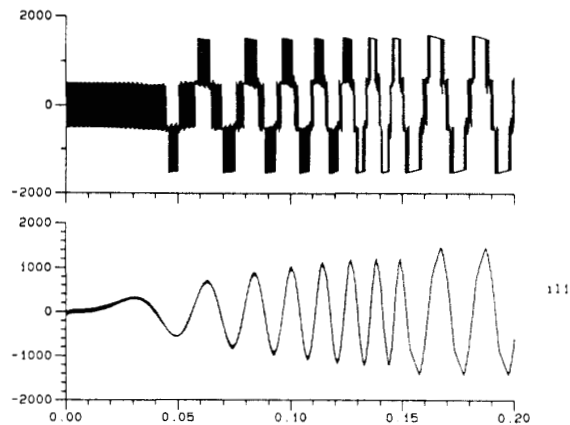
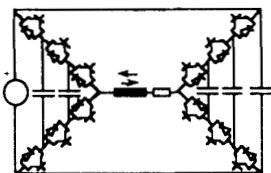


Figure 18
Over-modulation

- full-bridge configuration can be used to further improve the output voltage waveform by using two interleaved set of control signals (Fig. 19).



Full-bridge 7-level inverter

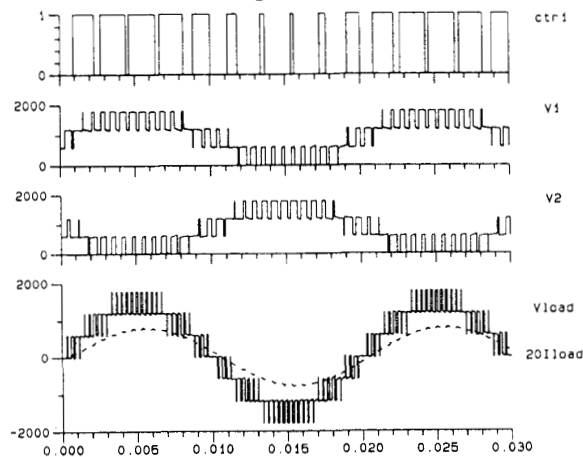
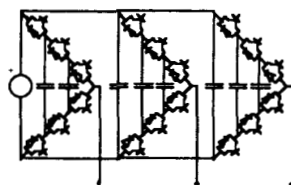


Figure 19
Full bridge operation

- three-phase configuration is possible (Figure 20).



Three-phase 7-level inverter

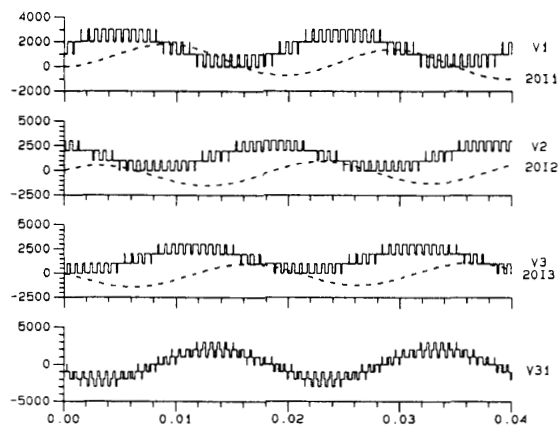


Figure 20
Three-phase operation

- unmodulated operation (switching frequency equal to the frequency of the AC current source) is possible. However, to cancel the average current in the capacitor, a different control strategy is required (Figure 21).

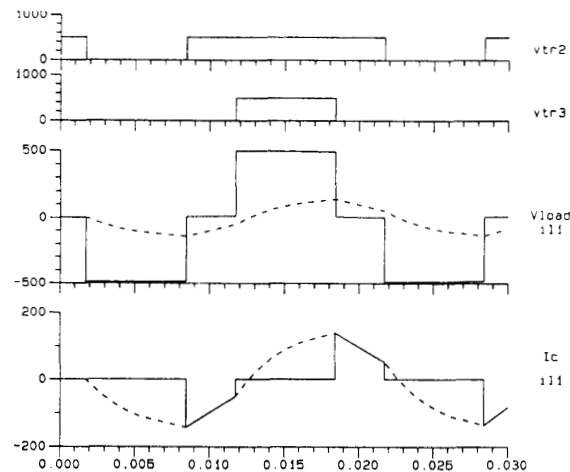


Figure 21
Unmodulated operation

CONCLUSION

It has been shown that in high voltage applications where series connection of switches is required, a new topology (ref 4) allows unprecedented improvements of the output harmonic spectrum. The control strategies proposed in this paper remain quite simple and their modularity allows straightforward generalization. Another advantage of this topology is to allow dV/dt limitation.

These characteristics are summarized in the following table.

	Series connection	Conventional 3-level	New multilevel
Voltage sharing	Difficult	Partial	Yes
dV/dt	Very high	Standard	Standard
Harmonic at f_s	High	#50%	Zero
Chopper	Yes	No	Yes
Inverter	Yes	Yes	Yes

Special fields of applications are high voltage power supplies with "high voltage" defined with respects to a given technology:

- GTO converters above 3kV (traction, induction heating,...)
- IGBT and MOSFET converters from 1000V (variable speed drives, high voltage-high frequency choppers,...)

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